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Steve Meisburger

Curriculum vitae

Education

Aug. 2014 **Doctor of Philosophy**, *Applied Physics, Cornell University*, Ithaca, NY.
Ion interactions with single- and double-stranded DNA measured using solution X-ray scattering.
Advisor: Lois Pollack

May 2007 **Bachelor of Arts**, *Physics, Carleton College*, Northfield, MN, *Summa cum laude*.

Appointments

2025 – **Research Associate**, *Chemistry & Chemical Biology, Cornell University*.

2022-2025 **Research Associate**, *Cornell High Energy Synchrotron Source, Cornell University*.

2020-2022 **Research Associate**, *Chemistry & Chemical Biology, Cornell University*.

2018-2020 **Postdoctoral Research Associate**, *Chemistry & Chemical Biology, Cornell University*.

2016-2018 **Postdoctoral Research Fellow**, *Chemistry, Princeton University*.

2014-2016 **Postdoctoral Research Associate**, *Chemistry, Princeton University*.

Awards and Honors

2016-2018 **NIH NRSA Postdoctoral Fellowship: "Allosteric motions of B_{12} enzymes"**, *Princeton University*, F32-GM117757. Sponsors: Nozomi Ando & Garnet Chan (Princeton Chemistry).

May 2017 **Best Poster: Technical Achievement**, *CHESS Users' Meeting*.

May 2017 **Travel Award**, *American Crystallographic Association*, Meeting in New Orleans, LA.

Apr. 2017 **Travel and Professional Development Award**, *American Society for Biochemistry and Molecular Biology*, Meeting in Chicago, IL.

2008-2011 **NIH Predoctoral Training Grant in Molecular Biophysics**, *Cornell University*, T32-GM008267.

May 2007 **Distinction awarded for Senior Integrative Exercise (Thesis)**, *Carleton College*.

Research Experience

2025– **Research Scientist**, *Cornell Chemistry & Chemical Biology*, Nozomi Ando's lab.

Project diffUSE, Astera Institute.
Multi-institution (Astera, Cornell, UCSF, Vanderbilt) open science project to enable accurate modeling of protein structural ensembles using diffuse scattering, the "hidden data" in X-ray crystallography. I lead the team in Nozomi Ando's group at Cornell and bring expertise in software, X-ray instrumentation, and data analysis to the collaboration.

2022-2025 **Beamline scientist**, *Cornell High Energy Synchrotron Source (CHESS)*.

Macromolecular crystallography at CHESS, NIH & NSF.
Supported international user program at beamline 7b2 (aka FlexX / X-BIO), designed X-ray instrumentation and data acquisition software (python GUI and backend), and pioneered avant-garde techniques including diamond anvil cell (DAC), diffuse scattering, and serial crystallography. Contributed to successful grant renewals for the NIH (MacCHESS) and NSF (CHEXS) programs.

2020-2022 **Researcher Scientist**, *Cornell Chemistry & Chemical Biology*, Nozomi Ando's lab.

Software and experimental methods for dynamic structural biology.
Developed methods to reveal protein motions from X-ray crystallography, SAXS, and cryo-EM data

2014-2020 **Postdoctoral Researcher**, *Princeton Chemistry, Cornell Chemistry & Chemical Biology*.
Advisor: Nozomi Ando

Protein dynamics from diffuse X-ray scattering.
Methods to use information between the Bragg peaks in a crystallography experiment (diffuse scattering) to determine correlated protein motions within the crystal. Collaborators: David Case (Rutgers), Sol Gruner (Cornell), Eddie Arnold (Rutgers).

Allosteric motions of B_{12} enzymes.
Investigated domain motions involved in regulating catalytic cycle of B_{12}-dependent enzymes Methionine Synthase and Corrinoid FeS Protein / Methyltransferase using solution scattering and non-conventional X-ray crystallography. Collaborator: Stephen Ragsdale (U. Michigan).

Allosteric regulation of phenylalanine hydroxylase.
Revealed role of domain motion in allosteric regulation by combining chromatography-coupled SAXS with mathematical deconvolution. Collaborator: Paul Fitzpatrick (UT Health San Antonio).

2008-2014 **Graduate Research Assistant**, *Cornell Applied & Engineering Physics*.
Advisor: Lois Pollack

Instrumentation and methods for time-resolved SAXS and cryo-SAXS.
Time-resolved SAXS experiments on RNA folding and nucleosome disassembly using microfluidic and stopped-flow mixers. Developed methods for cryo-cooling compatible with SAXS data collection. Collaborators: Ken Hampel (U. Vermont Medical Center), Rob Thorne (Cornell).

Unfolded states of nucleic acids and proteins.
Used SAXS to measure how single-stranded RNA and DNA homopolymers change shape in different salt solutions. Applied polymer theory and atomistic modeling to reconcile SAXS and single-molecule FRET data from protein-L and nucleic acids. Collaborators: Gilad Haran (Weizmann Institute), Kevin Plaxco (UCSB), Tobin Sosnick (U. Chicago).

Ion atmospheres around RNA and DNA.
Measured spatial distribution and number of positive ions surrounding DNA duplexes and RNA motifs using anomalous scattering (ASAXS), and compared results with continuum electrostatic theory and atomistic simulations. Collaborators: Kurt Andresen (Gettysburg College), David Case (Rutgers), Ron Elber (U.T. Austin).

2006 **Undergraduate Researcher**, *REU at the Santa Fe Institute*, Santa Fe, New Mexico.
Mentor: Alfred Hubler. Project: *Chaos in Wall Following Robots.*

2005 **Undergraduate Researcher**, *NNIN REU at Stanford University*, Palo Alto, California.
Mentors: Fabian Pease, Dan Pickard. Project: *Characterization of Beam Aberrations due to Magnetic Field Tilt in DIVA for Massively Parallel Electron Beam Lithography.*

Teaching Experience

Apr. 2021 **Instructor**, *High Pressure Biology Workshop*, Cornell High Energy Synchrotron Source, (Remote).

2020–2021 **Instructor**, *Everything BioSAXS: Getting started in biological small-angle x-ray solution scattering*, BioCAT beamline at Advanced Photon Source, (Remote).
Workshops in October 2020, March 2021

Aug. 2020 **Instructor**, *Applications of Small Angle Scattering to Structural Biology: An Introduction*, Workshop at the ACA Annual Meeting, (Remote).

2024–2018 **Instructor**, *BioSAXS Essentials: Getting Started in Biological Small-Angle X-ray Solution Scattering*, Cornell High Energy Synchrotron Source, Ithaca, NY.
Workshops in Nov. 2014, May 2016, May 2017, and April 2018

Oct. 2018 **Lecturer**, *9th annual SIBYLS BioSAXS workshop*, Advanced Light Source, Berkeley, CA.

Fall 2018 **Guest Lecturer**, *CHEM 2870: Introductory Physical Chemistry*, Cornell CCB.

Fall 2015 **Guest Lecturer**, *CHM 515: Biophysical Chemistry I*, Princeton Chemistry.

Spring 2014 **Guest Lecturer**, *CHM 516: Biophysical Chemistry II*, Princeton Chemistry.

2007-2008 **Teaching Assistant**, *AEP 3630: Electronic Circuits*, Cornell Applied & Eng. Physics.

Publications

[29] **Meisburger SP**, Ando N. *Scaling and merging macromolecular diffuse scattering with mdx2.* **Acta Crystallogr. D.** 80: 299-313 (2024).

[28] X Pei*, N Bhatt*, H Wang, N Ando*, **Meisburger SP***. *Introduction to diffuse scattering and data collection.* **Methods in Enzymology** 688: 1-42 (2023). *co-corresponding

[27] **Meisburger SP**, Ando N. *Processing macromolecular diffuse scattering data.* **Methods in Enzymology** 688: 43-86 (2023).

[26] **Meisburger SP**, Case D., Ando N. *Robust total X-ray scattering workflow to study correlated motion of proteins in crystals.* **Nat. Commun.** 14: 1228 (2023).

[25] Xu D*, **Meisburger SP***, Ando N. *Correlated motions in structural biology.* **Biochemistry** 60: 2331-40 (2021). *equal contribution

[24] **Meisburger SP***, Xu D*, Ando N. *REGALS: a general method to deconvolve X-ray scattering data from evolving mixtures.* **IUCrJ** 8: 225-237 (2021). *equal contribution

[23] **Meisburger SP**, Case DA, Ando N. *Diffuse X-ray Scattering from Correlated Motions in a Protein Crystal.* **Nat. Commun.** 11: 1271 (2020).

[22] Khan CA, **Meisburger SP**, Ando N, Fitzpatrick PF. *The phenylketonuria-associated substitution R68S converts phenylalanine hydroxylase to a constitutively active enzyme but reduces its stability.* **J. Biol. Chem.** 294(12): 4359-67 (2019).

[21] Parker MJ, Maggiolo AO, Thomas WC, Kim A, **Meisburger SP**, Ando N, Boal AK, Stubbe J. *An endogenous dAMP ligand in Bacillus subtilis class Ib RNR promotes assembly of a noncanonical dimer for regulation by dATP.* **PNAS** 115: E4594–E4603 (2018).

[20] **Meisburger SP***, Thomas WC*, Watkins MB*, and Ando N. *X-ray scattering studies of protein structural dynamics.* **Chem. Rev.** 117(12): 7615–72 (2017). *equal contribution

[19] **Meisburger SP** & Ando N. *Correlated motions from crystallography beyond diffraction.* **Acc. Chem. Res** 50: 580–583 (2017).

[18] Plumridge A, **Meisburger SP**, Andresen K, and Pollack L. *The impact of base stacking on the conformations and electrostatics of single-stranded DNA.* **Nucl. Acids Res.** 45: 3932-3943 (2017).

[17] Plumridge A*, **Meisburger SP***, Pollack L. *Visualizing single-stranded nucleic acids in solution.* **Nucl. Acids Res.** 45(9): e66 (2017). *equal contribution

[16] **Meisburger SP,** Taylor AB, Khan CA, Zhang S, Fitzpatrick PF, Ando N. *Domain movements upon activation of phenylalanine hydroxylase characterized by crystallography and chromatography-coupled small-angle X-ray scattering.* **JACS** 138(20): 6506–16 (2016).

[15] Chen Y, Tokuda JM, Topping T, **Meisburger SP**, Pabit SA, Gloss LM, Pollack L. *Asymmetric unwrapping of nucleosomal DNA propagates asymmetric opening and dissociation of the histone core.* **PNAS** 114: 334-339 (2016).

[14] Rustiguel JK, Soares ROS, **Meisburger SP**, Davis KM, Malzbender KL, Ando N, Dias-Baruffi M, Nonato MC. *Full-length model of the human galectin-4 and insights into dynamics of inter-domain communication.* **Sci. Rep.** 6: 33633 (2016).

[13] **Meisburger SP**, Pabit SA, Pollack L. *Determining the locations of ions and water around DNA from X-ray scattering measurements.* **Biophys. J.** 108: 2886-95 (2015).

[12] Hopkins JB, Katz AM, **Meisburger SP**, Warkentin MA, Thorne RE, Pollack L. *A microfabricated fixed path length silicon sample holder enables robust background subtraction for cryoSAXS.* **J. Appl. Cryst.** 48: 227-37 (2015).

[11] Chen Y, Tokuda JM, Topping T, Sutton JL, **Meisburger SP**, Pabit SA, Gloss LM, Pollack L. *Revealing transient structures of nucleosomes as DNA unwinds.* **Nucl. Acids Res.** 42: 8767-76 (2014).

[10] Nguyen HT, Pabit SA, **Meisburger SP**, Pollack L, Case DA. *Accurate small and wide angle X-ray scattering profiles from atomic models of proteins and nucleic acids.* **J. Chem. Phys.** 141: 22D508 (2014).

[9] **Meisburger SP**, Sutton JL, Chen H, Pabit SA, Kirmizialtin S, Elber R, Pollack L. *Polyelectrolyte properties of single stranded DNA measured using SAXS and single-molecule FRET: beyond the wormlike chain model.* **Biopolymers** 99: 1032-45 (2013).

[8] **Meisburger SP**, Warkentin M, Chen H, Hopkins JB, Gillilan RE, Pollack L, Thorne RE. *Breaking the radiation damage limit with Cryo-SAXS.* **Biophys. J.** 104(1): 227-36 (2013).

[7] Chen H*, **Meisburger SP***, Pabit SA, Sutton JL, Webb WW, Pollack L. *Ionic strength-dependent persistence lengths of single-stranded RNA and DNA.* **PNAS** 109(3): 799-804 (2012). *equal contribution

[6] Yoo TY*, **Meisburger SP***, Hinshaw J*, Pollack L, Haran G, Sosnick TR, Plaxco K. *Small-angle X-ray scattering and single-molecule FRET spectroscopy produce highly divergent views of the low-denaturant unfolded state.* **J. Mol. Biol.** 418(3-4): 226-36 (2012). *equal contribution

[5] Kirmizialtin S, Pabit SA, **Meisburger SP**, Pollack L, Elber R. *RNA and its ionic cloud: solution scattering experiments and atomically detailed simulations.* **Biophys. J.** 102(4): 819-28 (2012).

[4] Blose JM, Pabit SA, **Meisburger SP**, Li L, Jones CD, Pollack L. *Effects of a protecting osmolyte on the ion atmosphere surrounding DNA duplexes.* **Biochemistry** 50(40): 8540-7 (2011).

[3] Li L, Pabit SA, **Meisburger SP**, Pollack L. *Double-stranded RNA resists condensation.* **Phys. Rev. Lett.** 106: 108101 (2011).

[2] Pabit SA, **Meisburger SP**, Li L, Blose JM, Jones CD, Pollack L. *Counting ions around DNA with anomalous small-angle X-ray scattering.* **JACS** 132(46): 16334-6 (2010).

[1] Pabit SA, Qiu X, Lamb JS, Li L, **Meisburger SP**, Pollack L. *Both helix topology and counterion distribution contribute to the more effective charge screening in dsRNA compared with dsDNA.* **Nucl. Acids Res.** 37: 3887-96 (2009).

Patents

[1] **Meisburger SP**, Warkentin MA, Hopkins JB, Katz AM, Pollack L, and Thorne RE. (2018). *Apparatus and methods for low temperature small angle X-ray scattering.* US 9,927,336 (June 4, 2012).

Presentations: Talks

[21] 8th Ringberg Workshop on Structural Biology (with XFELs), February 7, 2025. (Invited)

[20] ACA Annual Meeting, Session 1.2.3: Extreme Biophysical & Biochemical Environments. July 8, 2024.

[19] SSRL/LCLS Usersâ Meeting, Application of SAXS in Structural Biology and Biophysics. September 28, 2023. (Invited)

[18] ACA Annual Meeting, Session 1.1.2 Data Analysis Software. July 8, 2023.

[17] CCP4 Study Weekend, Data - subtle details to big insights. Jan 6, 2023. (Invited)

[16] ACA Annual Meeting, Session 2.1.4 Macromolecular Structure Under Physiological Conditions. July 31, 2022.

[15] U. at Buffalo, Protein Science Group Seminar. (Remote) Oct. 6, 2021. (Invited)

[14] SSRL/LCLS Users' meeting. (Remote) Sep. 21, 2021. (Invited)

[13] Harvard-Hamburg Crystallographic Curiosities Seminar Series. (Remote) May 27, 2021. (Invited)

[12] SBGrid Weekly Webinar Series. (Remote) April 6, 2021. (Invited)

[11] ACA Annual Meeting, Session 2.1.4: Frontiers in SAS. (Remote) Aug. 3, 2020. (Invited)

[10] CHESS Commons Meeting. (Remote) Mar. 20, 2020. (Invited)

[9] ACS Northeast Regional Meeting. Saratoga Springs, NY. June 25, 2019. (Invited)

[8] Center for Nonlinear Studies, Los Alamos National Laboratory. Oct. 16, 2018. (Invited)

[7] ACA Annual Meeting, Session 1.1.1: Closing the R-Factor Gap in Protein Crystallography. Toronto, Ontario. July 21, 2018.

[6] ACA Annual Meeting, Session 1.1.1: Hybrid Methods - BioSAXS. New Orleans, LA. May 27, 2017.

[5] ASBMB Annual Meeting, Spotlight Session: Structural Dynamics of Enzymes. Chicago, IL. April 24, 2017.

[4] CHESS-U Workshop: Biomolecules in Motion. Ithaca, NY. June 8, 2016. (Invited)

[3] Biophysical Society Meeting, Platform: DNA Structure and Dynamics. San Francisco, CA. Feb. 18, 2014.

[2] CHESS Users' Meeting. Ithaca, NY. June 2013. (Invited)

[1] RNA-UNY. Rensselaerville, NY. Oct. 9, 2010.

Professional Memberships

2020– American Association for the Advancement of Science (AAAS)
2017– American Crystallographic Association (ACA)
2009-2015 Biophysical Society

Professional Activities

April 2025 Proposal review panel for LCLS run 25
July 2024 Session chair, ACA Meeting, *3.1.4: Data Analysis Software & Applications*
Sept. 2024 Session chair, 82st Pittsburgh Diffraction Conference, *2a: High Pressure Biology*
July 2023 Session chair, ACA Meeting, *4.2.3: Computatonal techniques for SAS*
Aug. 2021 Session chair, ACA Meeting, *BioWAXS: Experiment and Interpretation.*
2021 Chair, Small Angle Scattering Scientific Interest Group (SAS SIG) of the ACA
Aug. 2020 Session chair, ACA Meeting, *Structural Dynamics I: Protein collective motions studied by X-ray scattering and diffraction.*
2020 Chair-elect, SAS SIG of the ACA
Aug. 2019 Session chair, ACA Meeting, *Diffuse scattering for biological structure and dynamics.*
2016-2019 Elected member, CHESS User Executive Committee
Feb. 2014 Platform session chair, Biophysical Society Meeting
2025– Reviewer for *Biomacromolecules*
2021– Reviewer for *Structural Dynamics Journal*
2020– Reviewer for *Review of Scientific Instruments*
2017– Reviewer for *IUCrJ*
2015– Reviewer for *JACS*